Puffing Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Flaking / Puffing Machine

What it is used for

A flaking or puffing machine processes suitable grains or food materials using controlled heat, pressure or mechanical action. It produces flakes or puffed products for further preparation or consumption according to the machine design.

Main parts

- Identify the feed hopper or material loading opening.
- Locate the heating chamber, roasting section or pressure vessel, depending on design.
- Identify the rollers, dies or expansion chamber used to form the product.
- Locate the motor, drive mechanism and transmission components, where fitted.
- Identify the temperature, pressure and speed controls, where provided.
- Locate the discharge outlet and product collection tray.
- Identify protective guards, emergency stop and cleaning access points.

Before you start

1. Read the operating instructions and receive supervised training before use.
2. Inspect the frame, guards, electrical connections and moving parts for damage.
3. Confirm the machine is designed for the intended grain or food material.
4. Clean and dry food-contact surfaces and prepare suitable collection containers.
5. Check the heating system, pressure indicators and safety devices, where fitted.
6. Ensure ventilation and dust control are adequate for the material being processed.
7. Wear suitable protective clothing, hair covering and heat-resistant protection where required.

How to use

1. Check that the machine is clean, correctly assembled and ready for operation.
2. Select the appropriate operating settings for the material and product type.
3. Load the specified quantity of prepared grain or food material into the feed hopper.
4. Close all covers and ensure safety guards are correctly fitted.
5. Start the machine according to the manufacturer's procedure.
6. Allow the heating or processing system to reach the specified operating condition.
7. Feed material gradually and monitor the product formation and discharge.
8. Adjust feed rate or settings only as permitted by the operating instructions.
9. Stop feeding if abnormal noise, overheating, blockage or irregular product formation occurs.
10. Switch off and isolate power before opening the machine or removing trapped material.
11. Collect the finished product in a clean container and allow it to cool safely.

⚠ Safety

- Keep hands away from rollers, dies, hot surfaces and moving feed mechanisms.
- Never open a pressurised chamber or bypass pressure-relief devices.
- Use heat-resistant gloves and eye protection where hot material may splash or escape.
- Isolate power before clearing blockages, cleaning or adjusting internal components.
- Do not operate the machine with damaged guards or faulty safety controls.
- Follow the specified moisture, temperature and pressure limits for the machine and material.

Care & storage

- Disconnect power and allow hot components to cool before cleaning.
- Remove grain residue and food deposits using approved cleaning tools.
- Inspect rollers, dies, guards and seals for wear or damage.
- Maintain lubrication and service schedules according to manufacturer instructions.

Common mistakes to avoid

- Overloading the hopper and causing uneven processing or blockage.
- Using unsuitable moisture levels or settings for the selected grain.
- Opening hot or pressurised components before safe shutdown and cooling.
- Cleaning near moving parts or leaving food residue inside the machine.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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